Model Line Coverage Report

Verification date: 20 October 2026 (Asia/Hong_Kong).

Command: /Library/Frameworks/Python.framework/Versions/3.12/bin/python3.12 tools/report_model_coverage.py

Runtime: Python 3.12.6, macOS-15.7.3-arm64-arm-64bit.

Measured model unit tests: 175; all passed. Covered executable model lines: **659/661 (99.70%)**.

Method: standard-library trace line events, intersected with recursively compiled bytecode source-line tables (dis.findlinestarts). Each positive physical source line counts once. Blank/comment/function-docstring-only lines have no executable bytecode; module/class docstring assignments count when compiled as instructions. Imports are traced from a cold process. Every model .py file is inventoried, including completely unimported/unexecuted modules. No CLI/process/tool tests or CLI/tests/tools/standard-library source lines enter this measurement. Tracing is restored at each test boundary if an intentional recursion-limit test disabled the hook. Any missed lines within that test remain uncovered. This is line coverage, not branch coverage or requirements coverage.

Model file	Covered	Executable	Line coverage	Uncovered lines
model/__init__.py	1	1	100.00%	None
model/criteria.py	115	115	100.00%	None
model/errors.py	23	23	100.00%	None
model/persistence.py	117	117	100.00%	None
model/records.py	70	70	100.00%	None
model/search.py	158	159	99.37%	189
model/service.py	97	97	100.00%	None
model/validation.py	78	79	98.73%	16

This PDF is the separate formal Testing line-coverage report required by Project Description page 3, Deliverable 4.b, placed at the source root. It reports actual local execution; no Windows execution is claimed. The editable measured source remains MODEL_LINE_COVERAGE.md.

Source snapshot

- model/__init__.py: 81c189af13eb429d1fb802e3b185e08ad37b1b4de0a53c4f52c535b621bdfd98

- model/criteria.py: 4d3a9b181ab6726ff101b9fc738382b070ad88d9e91ca941b826fee49f1fecfc

- model/errors.py: 0cd3009c9412913940685531bfc89d2c025c215caaa548939e76ca21a1941b96

- model/persistence.py: 3b3bf96654f191f4a41470cca9ff8d390af00ebc6a151a3dcc67c8150fc519b6

- model/records.py: c9ff9029ddda1883378e19ea5130d9583e999a1cd0f097a10560d4ce6eb61b51

- model/search.py: dc395ed5d255bc6c5765b4ce957968786290000a9df09be492c75d02eedbf6b0

- model/service.py: b57f479884b34b5e6ad6e952ac491b5d10693921159d130f1e64067337a7f521

- model/validation.py: cfbaab2471b97b9f1ff95a86e1f787ec6813032cd0de22ebe8e8aa5352a85070

Independent accounting verification

Five coverage-tool tests pass: executed/partial/completely unexecuted modules; non-model and impossible-line hit exclusion; weighted aggregate arithmetic; failed-suite nonpublication; and trace-hook restoration for the next test. These tool tests are not in the model numerator or denominator.

An initial measurement undercounted later tests because deliberate recursion errors can disable CPython tracing. That instrumentation result was discarded. The validated runner restores tracing at each test boundary; missed lines within an interruption remain honestly uncovered. MODEL_LINE_COVERAGE.md and this PDF report the corrected actual measurement, not an inferred percentage.

The full application suite also passed 249 methods. Of those, only the 175 model methods were measured here. CLI/controller/view/process tests and the five accounting tests do not inflate model unit-test line coverage. Requirements coverage is separately reported in REQUIREMENTS_COVERAGE.pdf.